

Experiment 5: Implement Half and Full Subtractor

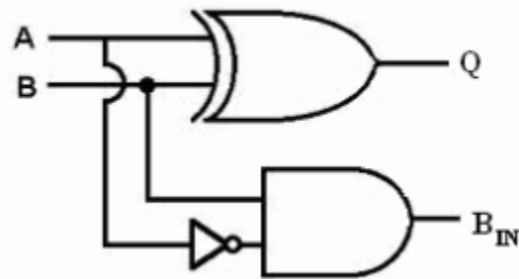
Theory:

In a similar fashion subtraction can be performed using binary numbers. The truth table for a single bit or half-subtractor with inputs A and B is given below along with its circuit diagram (. A full subtractor circuit accepts a minuend (A) and the subtrahend (B) and a borrow (BIN) as inputs from a previous circuit. A full subtractor circuit can be realized by combining two half subtractor circuits and an OR gate as shown

Circuit Diagram:

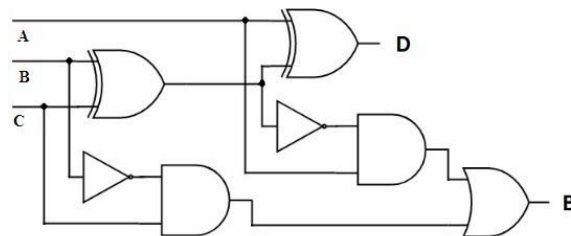
1-bit Subtractor with Borrow

A	B	Q	B _{IN}
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



1-bit Full Subtractor with B_{N-1} & B_N

B _{N-1}	A	B	Q	B _N
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	0
1	1	1	1	1



Experiment: Construct the truth table and match with the truth table